Won Rhim

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EXPERIENCE

Cubic May 2024 – Present
Software Engineer *Ashburn, VA*

- Designed and developed a cross-domain solutions device GUI application in C++ using the Qt Framework, enabling secure reading, modification, and persistence of configuration settings.
- Integrated the GUI with a configuration management API by implementing an adapter layer that bridged the GUI's data model requirements with existing data object classes without requiring modification of the underlying API.
- Engineered a role-based access control (RBAC) system that dynamically enables and disables UI widgets based on the authenticated user's role and permission set, enforcing least-privilege principles across the application.
- Developed a secure Linux user account management feature by implementing a suite of CLI tools that performed user provisioning and configuration under least-privilege access controls enforced via sudoers policy.
- Implemented a two-person integrity control workflow requiring users of different roles to approve configuration changes, leveraging session-based state management to track configuration change requests and enforce dual-authorization before applying modifications.

Hughes Network Systems Nov 2020 – Nov 2022
Software Engineer *Gaithersburg, MD*

- Developed a suite of multi-threaded microservices using Protobuf-based messaging to build the system controller for a satellite network gateway, performing network resource monitoring and management tasks with 20-millisecond precision.
- Owned the end-to-end design, development, and integration of two microservices into the existing system architecture.
- Drove the initial implementation of Helm charts for versioning microservices configuration on Kubernetes clusters, streamlining deployment setup and reducing manual effort for live integration tests.
- Refactored a network resource statistics microservice by creating a templatized Protobuf parser that generalized handling of multiple message types, reducing code duplication and improving extensibility for new message types.
- Transitioned the microservices suite from synchronous UDP message handling to an asynchronous Publish/Subscribe architecture using Redis, reducing per-message processing overhead by 20+ milliseconds.
- Developed a Python testing tool that restored microservices to a nominal state between automated regression test runs, eliminating the need for service restarts and significantly reducing total testing time.
- Continuously refined software design documents against evolving system requirements, driving alignment between acceptance criteria test results and implementation specifications.

Fraunhofer CESE May 2018 – Oct 2018
Software Engineer Intern *Riverdale, MD*

- Developed a cross-platform iOS/Android mobile gaming application for the University of Maryland's College of Medicine, Pharmacy, and Nursing using the MVC pattern and React Native.
- Designed and implemented a back-end REST API with Django, creating a hierarchical set of pathogen and disease models using object-oriented programming principles.
- Collaborated with front-end developers to design API endpoints and data formats, and built reusable React Native components for REST API integration.

EDUCATION

George Mason University May 2020
Bachelor of Science in Computer Science *Fairfax, VA*

SKILLS

Languages: C++, C, Java, Python, Ruby, HTML, CSS, JavaScript
Other: Boost C++ Libraries, CMake, Docker, Git, GoogleTest, Helm, Jenkins, Kubernetes, Linux, Protobuf, pytest, REST API development